



Figure 1: Image showing my creative assignment

This item is a handmade model representing fungi and their respective sexual spore producing structures from the two major phyla in the fungal kingdom, Basidiomycota and Ascomycota. More than 98% of described terrestrial fungi belong to these two phyla (Wang et al., 2010), with Ascomycota being the larger group. While there are other distinguishing features between Basidiomycota and Ascomycota, such as differences in life cycles and hyphal structures, sexual spore production is the most fundamental and easily observable characteristic for classification (OpenStax, 2018).

The purpose of this model is to visualise the mode of sexual spore production, a key distinguishing feature. Basidiomycota produce basidiospores on specialised structures called basidia, where the spores are formed externally on structures called sterigmata. Each basidium typically yields 4 basidiospores (Morrow & Fraser, 2009). In contrast, Ascomycota produce ascospores within sac-like structures known as asci, with each ascus typically containing eight ascospores (Bennett & Turgeon, 2016). Both the basidium and ascus are microscopic structures and are therefore represented in enlarged form in this model for clarity.

The three examples I have created for the Basidiomycota phylum represent common morphological forms: gills (*Chlorophyllum molybdites*), teeth (*Hydnum elatum*), and pores (*Earliella scabrosa*), illustrating the diversity of basidiomycete fruiting bodies whose surfaces bear basidia that produce basidiospores externally. In comparison, the three examples for the Ascomycota phylum represent common ascocarp forms: apothecia (*Cookeina speciosa*), perithecia (embedded within a stroma) (*Xylaria polymorpha*), and cleistothecia (*Tuber melanosporum*), illustrating the structural diversity of ascocarps, all of which produce ascospores internally within asci.

Citations:

Bennett, R. J., & Turgeon, B. G. (2016). Fungal sex: The Ascomycota. *Microbiology Spectrum*, 4(6). <https://doi.org/10.1128/microbiolspec.FUNK-0005-2016>

Morrow, C. A., & Fraser, J. A. (2009). *Sexual reproduction and dimorphism in the pathogenic basidiomycetes*. **FEMS Yeast Research**, 9(2), 161–177. <https://doi.org/10.1111/j.1567-1364.2008.00475.x>

OpenStax. (2018). Classifications of fungi. In *Biology 2e*. Rice University. https://bio.libretexts.org/Bookshelves/Introductory_and_General_Biology/General_Biology_1e_%28OpenStax%29/5%3A_Biological_Diversity/24%3A_Fungi/24.2%3A_Classifications_of_Fungi

Wang, H., Guo, S., Huang, M., Lumbsch, H. T., & Wei, J. (2010). *Ascomycota has a faster evolutionary rate and higher species diversity than Basidiomycota*. **Science China Life Sciences**, 53(10), 1163–1169. <https://doi.org/10.1007/s11427-010-4063-8>